DGMS Examination 2019 — Question 55 Detailed Solution

Junior Scientific Officer (Geology) | Plate Tectonics & Structural Geology

QUESTION 55

Along which of the following plate boundaries, the moving plates grind past each other without producing or destroying lithosphere?

(a) Diverging plate boundary

(b) Converging plate boundary only

(c) Transform fault plate boundary only CORRECT ANSWER

(d) Converging plate boundary and transform fault plate boundary both

1. Executive Summary & Answer Verification

The correct option is **(c) Transform fault plate boundary only**.

In plate tectonics theory (first formulated by J. Tuzo Wilson in 1965 in his seminal paper *"A New Class of Faults and their Bearing on Continental Drift"*), the lithospheric plates interact along three distinct classes of boundaries:

- **Divergent Boundary (Constructive):** Plates move away from one another. Upwelling asthenospheric mantle undergoes decompression melting to form new oceanic crust. Lithosphere is **produced** (created).
- **Convergent Boundary (Destructive):** Plates move toward one another. Denser oceanic lithosphere sinks into the mantle along Wadati-Benioff subduction zones. Lithosphere is **destroyed** (consumed).
- **Transform Fault Boundary (Conservative):** Plates slide horizontally past one another along strike-slip faults. Because plate motion is strictly parallel to the boundary, lithosphere is **neither created nor destroyed** ("conservative margin"). The plates merely grind past each other.

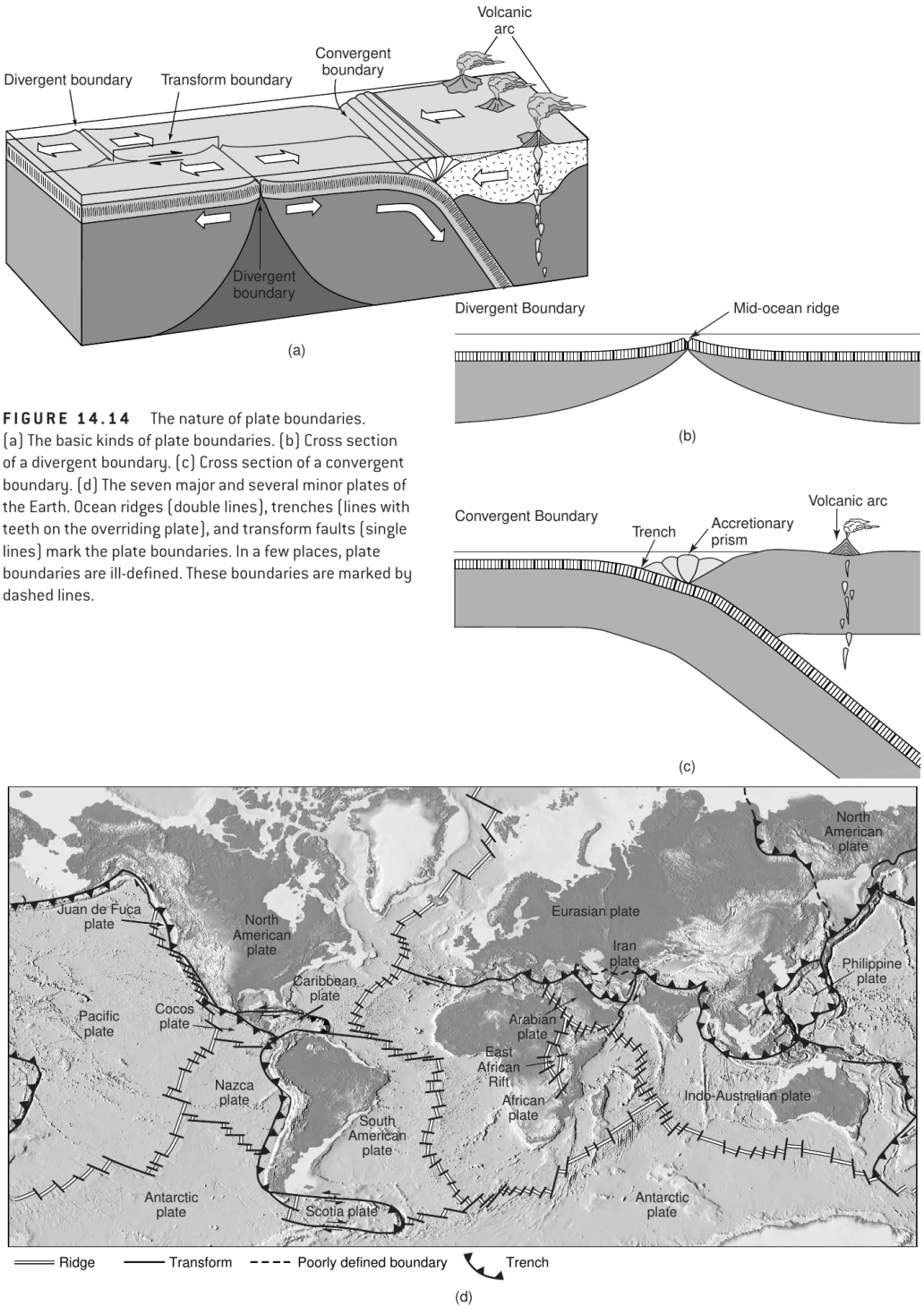


FIGURE 14.14 The nature of plate boundaries. (a) The basic kinds of plate boundaries. (b) Cross section of a divergent boundary. (c) Cross section of a convergent boundary. (d) The seven major and several minor plates of the Earth. Ocean ridges (double lines), trenches (lines with teeth on the overriding plate), and transform faults (single lines) mark the plate boundaries. In a few places, plate boundaries are ill-defined. These boundaries are marked by dashed lines.

Figure 55.1: *The three fundamental types of plate boundaries: (Left) Divergent Boundary (constructive, crust created); (Center)* ***Transform Boundary*** *(conservative, plates grind past each other with no creation or destruction of lithosphere); (Right) Convergent Boundary (destructive, crust subducted and destroyed). (Van der Pluijm & Marshak, Earth Structure, Figure 14.14, p. 357).*

2. Kinematics & Mechanics of Transform Boundaries

2.1 J. Tuzo Wilson's Conservative Principle

Before 1965, geologists viewed all strike-slip faults as standard transcurrent structures whose displacement terminated in regional distortion. Wilson recognized that **transform faults** are fundamentally different because they link other active plate boundaries (ridge-to-ridge, ridge-to-trench, or trench-to-trench):

- Along the active segment between two offset spreading ridge axes, plate motion is purely strike-slip. The relative velocity vector ($\mathbf{v}_{\text{rel}}$) is exactly parallel to the strike of the fault trace ($\mathbf{v} \parallel \hat{\mathbf{t}}$).
- Because there is no component of plate divergence ($\nabla \cdot \mathbf{v} = 0$) or plate convergence, the surface area of the Earth's lithosphere is strictly conserved:
  \frac{d A_{\text{lithosphere}}}{dt} = 0 (\text{Conservative Plate Margin})
- Beyond the active ridge crests, the transform fault grades into inactive **fracture zones**, where both sides belong to the same plate and travel in the same direction at equal speed.

2.2 Prominent Geological Examples

1. **Oceanic Transform Faults:** The Mid-Atlantic Ridge and East Pacific Rise are segmented every 50–100 km by oceanic transform faults (e.g., Romanche, Vema, and Kane fracture zones), accommodating the geometry of seafloor spreading on a spherical Earth.
2. **Continental Transform Faults:** Where plate motion cuts across continental landmasses:
   - **San Andreas Fault (California, USA):** Accommodates right-lateral strike-slip motion (~35 mm/yr) between the Pacific Plate and North American Plate.
   - **North Anatolian Fault (Turkey):** Accommodates westward extrusion of the Anatolian microplate relative to the Eurasian Plate.
   - **Dead Sea Transform (Middle East):** Connects Red Sea rifting to the Taurus collision zone.
   - **Alpine Fault (New Zealand):** Accommodates oblique strike-slip motion between the Pacific and Australian plates.

3. Systematic Comparison of Plate Boundary Regimes

Plate Boundary Type	Kinematic Motion	Tectonic Classification	Lithospheric Mass Budget	Associated Stress State	Volcanism & Seismicity
Divergent	Moving apart (separation)	**Constructive** margin	**Lithosphere CREATED** (~3.4 km²/yr globally)	Tensional (σ_3 horizontal)	Abundant basaltic volcanism; shallow normal earthquakes
Convergent	Moving together (collision)	**Destructive** margin	**Lithosphere DESTROYED** (consumed by subduction)	Compressional (σ_1 horizontal)	Explosive andesitic volcanism; deep to shallow earthquakes (Benioff)
				Shear stress (σ_1 & $	**No volcanism** (volcanism-free);

Plate Boundary Type	Kinematic Motion	Tectonic Classification	Lithospheric Mass Budget	Associated Stress State	Volcanism & Seismicity
Transform Fault	**Grind past each other (lateral shear)**	**Conservative** margin	**NEITHER created NOR destroyed**	\sigma_3$ horizontal at 45°)	severe shallow strike-slip earthquakes

4. Evaluation of All Options

- **(a) Diverging plate boundary — INCORRECT:** Divergent margins continuously generate new lithosphere via seafloor spreading.
- **(b) Converging plate boundary only — INCORRECT:** Convergent boundaries actively consume and destroy lithosphere into the asthenosphere.
- **(c) Transform fault plate boundary only — CORRECT:** Exactly satisfies the conservative criteria: plates grind past each other with purely horizontal strike-slip motion without creating or destroying lithospheric plates.
- **(d) Converging plate boundary and transform fault plate boundary both — INCORRECT:** Converging boundaries destroy vast quantities of lithosphere, invalidating this option.

Textbook Authority

Ben A. van der Pluijm & Stephen Marshak, *Earth Structure* (2nd ed., Chapter 14: The Crust and Moho, p. 356):
"At transform boundaries, plates slide past one another along strike-slip faults. Because no new lithosphere is created and none is destroyed at transform boundaries, they are sometimes called conservative plate boundaries."
J. Tuzo Wilson (1965), *A New Class of Faults and their Bearing on Continental Drift* (Nature):
"Transform faults are conservative plate boundaries where moving lithospheric plates slide horizontally past one another, with neither crustal generation nor consumption."